Bridging the Language Gaps in Large Language Models with Inference-Time Cross-Lingual Intervention

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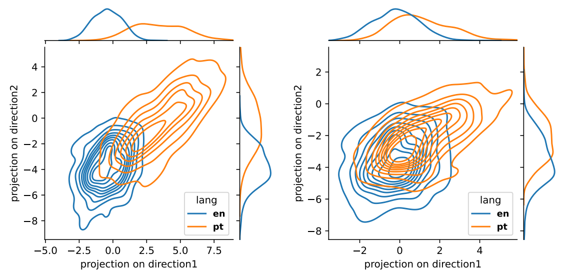
Abstract

Large Language Models (LLMs) have shown remarkable capabilities in natural language processing but exhibit significant performance gaps among different languages. Most existing approaches to address these disparities rely on pretraining or fine-tuning, which are resource-intensive. To overcome these limitations without incurring significant costs, we propose **Inference-Time Cross-Lingual Intervention (INCLINE)**, a novel framework that enhances LLM performance on low-performing (source) languages by aligning their internal representations with those of high-performing (target) languages during inference. INCLINE initially learns alignment matrices using parallel sentences from source and target languages through a Least-Squares optimization, and then applies these matrices during inference to transform the low-performing language representations toward the high-performing language space. Extensive experiments on nine benchmarks with five LLMs demonstrate that INCLINE significantly improves performance across diverse tasks and languages, compared to recent strong baselines. Our analysis demonstrates that INCLINE is highly cost-effective and applicable to a wide range of applications. In addition, we release the code to foster research along this line.[1]



(a) Before intervention (b) After intervention

Figure 1: Bivariate kernel density estimation plots displaying the representations (hidden states of the last token) from 100 random examples in English (blue) and their Portuguese translations (orange) from XCOPA (Ponti et al., 2020). After intervention using INCLINE, the Portuguese representations are aligned closer to the English representations.

1 Introduction

Large Language Models (LLMs) have achieved remarkable success across a variety of natural language processing tasks, demonstrating strong capabilities in language understanding and generation (OpenAI, 2023; Dubey et al., 2024; Mesnard et al., 2024; Anthropic, 2024; OpenAI, 2024a,b). However, despite these advancements, most state-of-the-art LLMs remain predominantly English-centric, exhibiting significant performance gaps among different languages (Petrov et al., 2023; Kumar et al., 2024), which can adversely affect user experience and potentially exclude large portions of the global population from accessing advanced AI services (Lai et al., 2023a; Wang et al., 2024a).

Addressing the performance gaps across languages is highly challenging. Recent approaches are mostly data-driven, such as multilingual supervised fine-tuning or continued pre-training (Üstün et al., 2024; Cui et al., 2023; Kuulmets et al., 2024). However, collecting and annotating large-scale datasets for numerous underrepresented languages is both time-consuming and resource-intensive (Xue et al., 2021; Lai et al., 2023b). Furthermore, training LLMs on multilingual data requires substantial computational resources, limiting their practicality for widespread applications, especially in resource-constrained settings (Muennighoff et al., 2023; Li et al., 2023a). Given these limitations, a natural question arises: *How can we bridge the performance gaps between high-performing and low-performing languages without incurring prohibitive costs?*

Inspired by Lample et al. (2018) showing that

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[1] https://github.com/weixuan-wang123/INCLINE

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通过推理时跨语言干预弥合大语言模型中的语言差距

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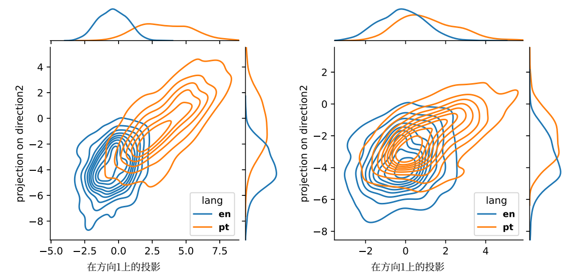
摘要

大语言模型（LLMs）在自然语言处理中展现出卓越能力，但在不同语言间存在显著的性能差距。大多数现有解决这些差异的方法依赖于预训练或微调，这些方法资源密集。为了在不产生显著成本的情况下克服这些局限性，我们提出**推理时跨语言干预（INCLINE）**，这是一个新颖的框架，通过在推理过程中将低性能（源语言）的内部表示与高性能（目标语言）的对齐，来提升大语言模型在低性能语言上的表现。INCLINE首先通过最小二乘优化，利用源语言和目标语言的平行句子学习对齐矩阵，然后在推理时应用这些矩阵，将低性能语言表示转换到高性能语言空间。在五个大语言模型上对九个基准进行的广泛实验表明，与近期强大的基线相比，INCLINE显著提升了跨多种任务和语言的性能。我们的分析表明，INCLINE具有很高的成本效益，并适用于广泛的应用场景。此外，我们发布了代码以促进这一方向的研究。[1]



(a) 干预前 (b) 干预后

图1：二元核密度估计图展示了来自XCOPA(Ponti等人，2020)的100个随机英语示例（蓝色）及其葡萄牙语翻译（橙色）的表示（最后一个词元的隐藏状态）。在使用INCLINE进行干预后，葡萄牙语的表示与英语的表示对齐得更紧密。

1 引言

大语言模型（LLMs）在各种自然语言处理任务中取得了显著成功，展现出强大的语言理解和生成能力（OpenAI，2023；Dubey等人，2024；Mesnard等人，2024；Anthropic，2024；OpenAI，2024a，b）。然而，尽管取得了这些进展，大多数最先进的大语言模型仍然主要以英语为中心，在不同语言之间表现出显著的性能差距（Petrov等人，2023；Kumar等人，2024），这可能会对用户体验产生不利影响，并可能使全球大部分人口无法访问先进的人工智能服务（Lai等人，2023a；Wang等人，2024a）。

解决跨语言性能差距极具挑战性。最近的方法大多是数据驱动的，例如多语言监督微调或持续预训练(Üstün等人，2024；Cui等人，2023；Kuulmets等人，2024)。然而，为众多代表性不足的语言收集和标注大规模数据集既耗时又耗费资源(Xue等人，2021；Lai等人，2023b)。此外，在多语言数据上训练大语言模型需要大量计算资源，限制了其在广泛应用中的实用性，尤其是在资源受限的环境中(Muen- nighoff等人，2023；Li等人，2023a)。鉴于这些局限性，一个自然的问题随之产生：我们如何在不产生过高成本的情况下，弥合高性能语言与低性能语言之间的性能差距？

受Lample等人(2018)的研究启发，该研究表明

[†]同等贡献。[1]
https://github.com/weixuan-wang123/INCLINE

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word embeddings in different languages can be aligned to a shared representation space through learned rotations for word translation, we propose **Inference-Time Cross-Lingual Intervention (INCLINE)**. This novel framework utilizes a group of learned alignment matrices that transform the representations (e.g., hidden states) of a low-performing (source) language into those of a high-performing (target) language during inference. Our framework comprises two main steps. First, we train the alignment matrices for each layer of LLM using parallel sentences from the source and target languages. The learning process is formulated as a Least-Squares optimization problem, where these alignment matrices are learned by minimizing the distance between the projected source language representations and their corresponding target language representations, without the need for extensive retraining or fine-tuning the LLM. Second, we apply the learned alignment matrices to transform the source language input representations into the target language representation space at each layer during inference. By integrating these steps, INCLINE leverages the rich representations learned from high-performing languages to enhance performance on downstream tasks involving low-performing languages. As shown in Figure 1, INCLINE effectively aligns the input representations in Portuguese to their parallel representations in English.

In this study, we conduct extensive experiments to validate the effectiveness of INCLINE on nine widely used benchmarks using five LLMs. Our results demonstrate that aligning internal representations using INCLINE significantly improves performance on diverse tasks among languages.

Our contributions are summarized as follows:

- We propose INCLINE, a cross-lingual intervention approach that enhances LLMs by transforming source language representations into a target language representation space during inference without requiring additional training of LLMs (see Section 3).
- We conduct extensive evaluations across five discriminative tasks and four generative tasks, covering 21 languages. Our experimental results show that INCLINE significantly improves model performance, boosting average accuracy by up to +4.96 compared to strong baselines (see Section 4).
- Our detailed analysis indicates that INCLINE is highly cost-effective, as it requires minimal computational resources while delivering substantial performance improvements (see Section 5). Moreover, we demonstrate that INCLINE is effective with regard to LLM backbones, model sizes, and in-context learning, underscoring its general applicability and potential for broader use in enhancing LLMs for underrepresented languages (see Section 6).

2 Related Work

Multilingual LLMs LLMs are pivotal in multilingual NLP tasks, typically leveraging external parallel datasets for training (Xue et al., 2021; Muennighoff et al., 2023; Chung et al., 2024; Li et al., 2025). For low-resource languages, data augmentation techniques generate parallel data by mining sentence pairs or translating monolingual text using machine translation tools (Edunov et al., 2018; Zhao et al., 2021; Ranaldi et al., 2023). However, these methods heavily rely on robust parallel corpora. To reduce data costs, studies have shifted toward Parameter-Efficient Fine-Tuning (PEFT) techniques (Pfeiffer et al., 2020; Parović et al., 2022; Agrawal et al., 2023; Wu et al., 2024; Wang et al., 2025) and cross-lingual embeddings mapping methods (Mikolov et al., 2013; Ormazabal et al., 2019; Wang et al., 2022), which still demand considerable computational resources.

Multilingual Prompting There is a growing interest in methods that do not require parameter adjustments. Prompting techniques have emerged, utilizing LLMs with multilingual prompts (Lin et al., 2021c, 2022; Shi et al., 2022b; Huang et al., 2023). However, these strategies face challenges like poor translation quality and prompt framing interference (Wang et al., 2024c). Additionally, their effectiveness varies by task, as recent research indicates that few-shot learning may not outperform zero-shot learning in translation tasks (Hendy et al., 2023; Wang et al., 2024d).

Intervention To address these challenges, we explore inference-time intervention techniques as cost-effective and efficient alternatives to traditional fine-tuning. Prior research in style transfer (Subramani et al., 2022; Turner et al., 2023), knowledge editing (Meng et al., 2022), and truthfulness shifting (Li et al., 2023b; Rimsky et al., 2024; Wang et al., 2024e) demonstrates the potential of linear probe-based interventions. However, these methods have been largely limited to mono-

不同语言的词嵌入可以通过学习到的旋转进行对齐，形成一个共享表示空间，以实现词语翻译。基于此，我们提出了**推理时跨语言干预（INCLINE）**。这一新颖框架利用一组学习到的对齐矩阵，在推理过程中将低性能（源）语言的表示（例如隐藏状态）转换为高性能（目标）语言的表示。我们的框架包含两个主要步骤。首先，我们使用源语言和目标语言的平行句，为大语言模型的每一层训练对齐矩阵。学习过程被表述为一个最小二乘优化问题，通过最小化投影后的源语言表示与其对应的目标语言表示之间的距离来学习这些对齐矩阵，无需对大语言模型进行大量的重新训练或微调。其次，在推理过程中，我们在每一层应用学习到的对齐矩阵，将源语言输入表示转换到目标语言表示空间。通过整合这些步骤，INCLINE利用从高性能语言学习到的丰富表示，来提升涉及低性能语言的下游任务性能。如图1所示，INCLINE有效地将葡萄牙语的输入表示与其在英语中的平行表示对齐。

在本研究中，我们进行了广泛的实验，使用五种大语言模型在九个广泛使用的基准上验证INCLINE的有效性。我们的结果表明，使用INCLINE对齐内部表示能显著提升跨语言多样任务上的性能。

我们的贡献总结如下：

- 我们提出了INCLINE，一种跨语言干预方法，它通过在推理过程中将源语言表示转换为目标语言表示空间来增强大语言模型，而无需对大语言模型进行额外训练（参见第3节）。
- 我们在五个判别性任务和四个生成式任务上进行了广泛的评估，覆盖21种语言。我们的实验结果表明，INCLINE显著提升了模型性能，与强大的基线相比，平均准确率最高提升了 +4.96（参见第4节）。
- 我们的详细分析表明，INCLINE具有极高的成本效益，因为它只需要极少的计算资源，却能带来显著的性能提升（参见第5节）。此外，我们证明INCLINE在大语言模型骨干架构、模型大小和上下文学习方面均有效，这突显了其广泛的适用性以及在增强针对代表性不足语言的大语言模型方面的巨大潜力（参见第6节）。

2 相关工作

多语言大语言模型 大语言模型在多语言自然语言处理任务中至关重要，通常利用外部平行数据集进行训练（Xue等人，2021；Muennighoff等人，2023；Chung等人，2024；Li等人，2025）。对于低资源语言，数据增强技术通过挖掘句子对或使用机器翻译工具翻译单语文本来生成平行数据（Edunov等人，2018；Zhao等人，2021；Ranaldi等人，2023）。然而，这些方法严重依赖于稳健的平行语料库。为了降低数据成本，研究已转向参数高效微调技术（Pfeiffer等人，2020；Parovi´c等人，2022；Agrawal等人，2023；Wu等人，2024；Wang等人，2025）和跨语言嵌入映射方法（Mikolov等人，2013；Ormazabal等人，2019；Wang等人，2022），这些方法仍需要大量的计算资源。

多语言提示人们对无需调整参数的方法越来越感兴趣。提示技术已经出现，利用大语言模型进行多语言提示（Lin等人,2021c, 2022; Shi等人, 2022b; Huang等人, 2023）。然而，这些策略面临着翻译质量差和提示框架干扰等挑战（Wang等人,2024c）。此外，其有效性因任务而异，因为最近的研究表明，在翻译任务中，少样本学习可能并不优于零样本学习（Hendy等人, 2023;Wang等人, 2024d）。

干预为应对这些挑战，我们探索推理时干预技术，将其作为传统微调的一种成本效益高且高效的替代方案。先前在风格迁移（Subramani等人，2022；Turner等人，2023）、知识编辑（Meng等人，2022）和真实性转变（Li等人，2023b；Rimsky等人，2024；Wang等人，2024e）方面的研究展示了基于线性探测的干预的潜力。然而，这些方法在很大程度上局限于单语

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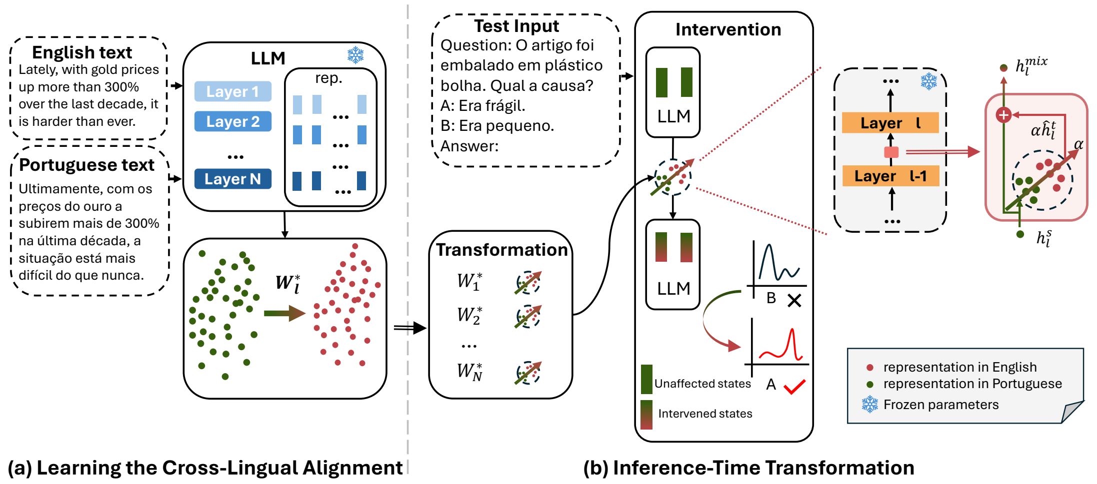


Figure 2: Framework of INCLINE. INCLINE involves two steps: (a) Learning the Cross-Lingual Alignment: sentence representations from a parallel dataset are used to train alignment matrices that map source (Portuguese) representations to the target (English) representations. (b) Inference-Time Transformation: this step adapts the source representations from downstream tasks into the target representation space using the alignment matrices.

lingual contexts. Our goal is to design a novel cross-lingual inference-time intervention that effectively aligns representations across languages, aiming to improve performance across multiple languages.

3 Methodology

In Figure 2, we illustrate the framework of INCLINE, which enhances LLMs through inference-time cross-lingual intervention. Our approach comprises two main steps:

- **Learning the Cross-Lingual Alignment**: Using parallel corpora, we train alignment matrices for each layer to map source language representations to target language representations (see Section 3.1).
- **Inference-Time Transformation**: During inference, we utilize the learned alignment matrices to transform input representations from the source language into the target language representation space, thereby improving the LLM's performance on tasks in the source language (see Section 3.2).

By minimizing the distance between the source language representations and their corresponding target language representations, we effectively reduce cross-lingual representation gaps and align representation spaces across languages.

3.1 Learning the Cross-Lingual Alignment

Inspired by Schuster et al. (2019) that align embeddings across languages with learned linear transformations, we aim to learn a cross-lingual alignment matrix $\boldsymbol{W}_l$ that aligns sentence representations from the source language to the target language at the l-th layer of LLM. Given a parallel dataset $D = \{(\boldsymbol{x}_i^{\mathrm{s}}, \boldsymbol{x}_i^{\mathrm{t}})\}_{i=1}^{N}$, where each $\boldsymbol{x}_i^{\mathrm{s}}$ is the i-th source sentence and $\boldsymbol{x}_i^{\mathrm{t}}$ is its corresponding translation in the target language. Both $\boldsymbol{x}_i^{\mathrm{s}}$ and $\boldsymbol{x}_i^{\mathrm{t}}$ are sequences of tokens. From these sequences, we extract sentence representations by taking the hidden state of the last token in each sequence, denoted as $\boldsymbol{h}_{i,l}^{\mathrm{s}} \in \mathbb{R}^d$ and $\boldsymbol{h}_{i,l}^{\mathrm{t}} \in \mathbb{R}^d$ for the source and target sentence, respectively, where d is the dimensionality of the hidden states.

To minimize the difference between the projected source sentence representations and the target sentence representations, our objective can be defined as a Least-Squares optimization problem:

$$\boldsymbol{W}_l^* = \underset{\boldsymbol{W}_l}{\operatorname{argmin}} \sum_{i=1}^{N} \left\| \boldsymbol{W}_l \boldsymbol{h}_{i,l}^{\mathrm{s}} - \boldsymbol{h}_{i,l}^{\mathrm{t}} \right\|^2 \quad (1)$$

This problem seeks the optimal $\boldsymbol{W}_l^*$ that aligns the source representations with the target representations by minimizing the distance between them. Hence, the closed-form solution to this optimization problem is:

$$\boldsymbol{W}_l^* = \left(\sum_{i=1}^{N} (\boldsymbol{h}_{i,l}^{\mathrm{s}})^{\top} \boldsymbol{h}_{i,l}^{\mathrm{s}} \right)^{-1} \left(\sum_{i=1}^{N} (\boldsymbol{h}_{i,l}^{\mathrm{s}})^{\top} \boldsymbol{h}_{i,l}^{\mathrm{t}} \right) \quad (2)$$

By applying the learned alignment matrix $\boldsymbol{W}_l^*$ to the source sentence representations, we effectively

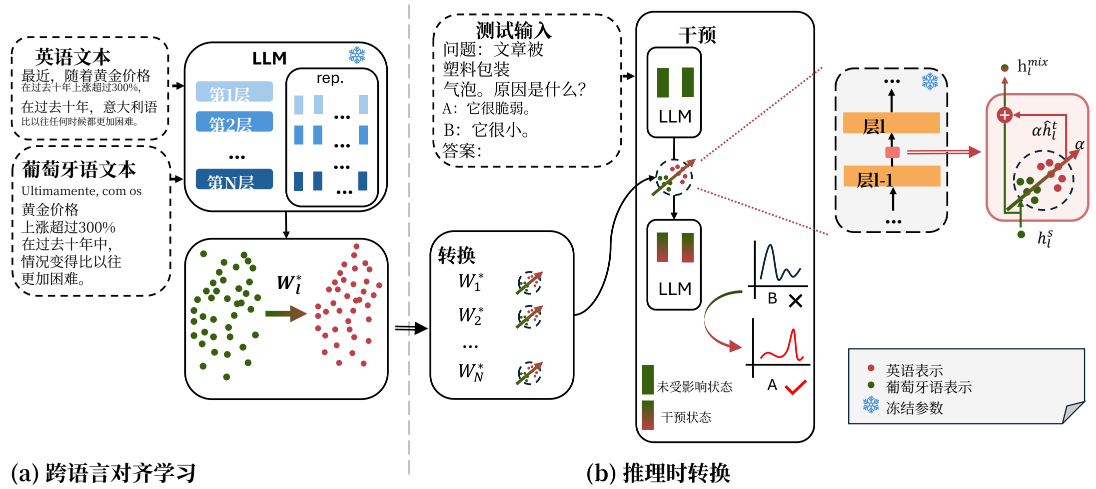


图2：INCLINE框架。INCLINE包含两个步骤：(a) 跨语言对齐学习：使用平行数据集中的语句表示来训练对齐矩阵，这些矩阵将源语言（葡萄牙语）表示映射到目标语言（英语）表示。(b) 推理时转换：此步骤使用对齐矩阵将下游任务中的源表示适配到目标表示空间。

语境。我们的目标是设计一种新颖的跨语言推理时干预，能有效对齐跨语言的表示，旨在提升在多种语言上的性能。

3 方法论

在图2中，我们阐述了IN-CLINE的框架，该框架通过推理时跨语言干预来增强大语言模型。我们的方法包含两个主要步骤：

- **跨语言对齐学习**：使用平行语料库，我们为每一层训练对齐矩阵，以将源语言表示映射到目标语言表示（参见第3.1节）。
- **推理时转换**：在推理过程中，我们利用学习到的对齐矩阵将输入表示从源语言转换到目标语言表示空间，从而提升大语言模型在源语言任务上的性能（参见第3.2节）。

通过最小化源语言表示与其对应目标语言表示之间的距离，我们有效减少了跨语言表示差距，并实现了跨语言的表示空间对齐。

3.1 跨语言对齐学习

受Schuster等人(2019年)的启发，他们通过学习的线性变换来对齐跨语言嵌入，我们的目标是学习一个跨语言对齐矩阵 $\boldsymbol{W}_l$，该矩阵将源语言的语句表示对齐到目标语言在LLM的第 l层。给定一个平行数据集 $D = \{(\boldsymbol{x}_i^{\mathrm{s}}, \boldsymbol{x}_i^{\mathrm{t}})\}_{i=1}^{N}$，其中每个 $\boldsymbol{x}_i^{\mathrm{s}}$ 是第i个源语句，$\boldsymbol{x}_i^{\mathrm{t}}$ 是其对应的目标语言翻译。$\boldsymbol{x}_i^{\mathrm{s}}$ 和 $\boldsymbol{x}_i^{\mathrm{t}}$都是词元序列。从这些序列中，我们通过获取每个序列中最后一个词元的隐藏状态来提取语句表示，分别表示为源语句的$\boldsymbol{h}_{i,l}^{\mathrm{s}} \in \mathbb{R}^d$ 和目标语句的$\boldsymbol{h}_{i,l}^{\mathrm{t}} \in \mathbb{R}^d$，其中 d 是隐藏状态的维度。

为了最小化投影后的源语句表示与目标语句表示之间的差异，我们的目标可以定义为一个最小二乘优化问题：

$$\boldsymbol{W}_l^* = \underset{\boldsymbol{W}_l}{\operatorname{argmin}} \sum_{i=1}^{N} \left\| \boldsymbol{W}_l \boldsymbol{h}_{i,l}^{\mathrm{s}} - \boldsymbol{h}_{i,l}^{\mathrm{t}} \right\|^2 \quad (1)$$

该问题旨在寻找最优的 $\boldsymbol{W}_l^*$，通过最小化源表示与目标表示之间的距离来对齐它们。因此，该优化问题的闭式解为：

$$\boldsymbol{W}_l^* = \left(\sum_{i=1}^{N} (\boldsymbol{h}_{i,l}^{\mathrm{s}})^{\top} \boldsymbol{h}_{i,l}^{\mathrm{s}} \right)^{-1} \left(\sum_{i=1}^{N} (\boldsymbol{h}_{i,l}^{\mathrm{s}})^{\top} \boldsymbol{h}_{i,l}^{\mathrm{t}} \right) \quad (2)$$

通过将习得的对齐矩阵 $\boldsymbol{W}_l^*$应用于源语句表示，我们有效地

map them into the target language's representation space. This alignment reduces cross-lingual representation discrepancies, allowing the model to leverage knowledge from the target language to improve performance on tasks in the source language.

3.2 Inference-Time Transformation

With the learned alignment matrix $\boldsymbol{W}_l^*$, we can enhance the LLM's processing of source language inputs by transforming their representations to the target representation space during inference.

We denote the hidden state of the last token of the test input $\boldsymbol{q}^s$ in the source language at the l-th layer of the LLM as $\boldsymbol{h}^{\mathrm{s}}_{q,l}$ and then project this source language representation into the target representation space using the alignment matrix $\boldsymbol{W}_l^*$:

$$\hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l} = \boldsymbol{W}_l^* \boldsymbol{h}^{\mathrm{s}}_{q,l} \tag{3}$$

To perform the cross-lingual intervention at the l-th layer using the intervention vector $\hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l}$, we adjust the original hidden state in source language $\boldsymbol{h}^{\mathrm{s}}_{q,l}$ by blending it with the projected hidden state in target language $\hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l}$. This adjustment is controlled by a hyperparameter α, which balances the influence between the source and target hidden states:

$$\boldsymbol{h}^{\mathrm{mix}}_{q,l} = \boldsymbol{h}^{\mathrm{s}}_{q,l} + \alpha \hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l} \tag{4}$$

Here, Equation 4 represents a shift of representation of source language towards target language representation by a magnitude of α times.

Decoding with Minimal Intervention In this work, we only conduct one single intervention on the last token of $\boldsymbol{q}^s$ by replacing $\boldsymbol{h}^{\mathrm{s}}_{q,l}$ with $\boldsymbol{h}^{\mathrm{mix}}_{q,l}$ for the test input $\boldsymbol{q}^s$ at the l-th layer of LLM. In such a way, we can effectively intervene the model output while preserve the features in the source language.

Comparison with ITI and CAA Recently, ITI (Li et al., 2023b) and CAA (Rimsky et al., 2024) have been proposed as interventions in the model behaviors by manipulating the selected attention heads and hidden states, respectively. INCLINE is distinct from ITI and CAA due to three primary differences. Firstly, ITI and CAA utilize a learned *static* intervention vector to alter model behaviors, whereas INCLINE leverages a set of alignment matrices to *dynamically* align input representations from the source language to the target language. Secondly, ITI and CAA apply the intervention vector across all token positions following the instruction, potentially causing excessive perturbation during inference. In contrast, INCLINE performs a single intervention solely on the last token of the input. Additionally, unlike ITI and CAA, which target on only a limited number of layers, INCLINE modifies the representations across all layers. These modifications enable the LLMs to comprehensively leverage their target language capabilities for multilingual prediction.

4 Experiments

In this section, we introduce our experimental setup (Section 4.1) and present our results in Section 4.2.

4.1 Experimental Setup

We present our evaluation tasks, model backbones, implementation details of INCLINE, and baselines in this section.

Evaluation Tasks We conduct extensive evaluations across nine diverse downstream tasks, categorized into two groups:

- **Discriminative Tasks**: XCOPA (Ponti et al., 2020), XStoryCloze (Lin et al., 2021b), XWinograd (Lin et al., 2021b), XCSQA (Lin et al., 2021a), XNLI (Conneau et al., 2018);
- **Generative Tasks**: MZsRE (Wang et al., 2024b), Flores (Goyal et al., 2021), WMT23 (Kocmi et al., 2023), MGSM (Shi et al., 2022a).

These tasks covers 21 languages including English (en), Arabic (ar), German (de), Greek (el), Spanish (es), Estonian (et), French (fr), Hindi (hi), Indonesian (id), Italian (it), Japanese (ja), Dutch (nl), Portuguese (pt), Russian (ru), Swahili (sw), Tamil (ta), Thai (th), Turkish (tr), Ukrainian (uk), Vietnamese (vi), and Chinese (zh). We include more details of these tasks in Appendix A.

Model Backbones In this work, we mainly use BLOOMZ-7B1-MT as our model backbone for all the baseline approaches, unless otherwise specified. To demonstrate the effectiveness of INCLINE across various model backbones, we include four additional LLMs: LLAMA3.1-8B-INSTRUCT (Dubey et al., 2024), LLAMA2-7B-CHAT (Touvron et al., 2023), MISTRAL-7B-INSTRUCT (Jiang et al., 2023), FALCON-7B-INSTRUCT (Almazrouei et al., 2023). We present these results in Section 6. For the MGSM task, we employ the MATHOCTOPUS (Chen et al., 2023),[2]

[2]https://huggingface.co/Mathoctopus/Parallel_7B

将其映射到目标语言的表示空间中。这种对齐减少了跨语言表示差异，使模型能够利用目标语言的知识来提高源语言任务的性能。

3.2 推理时转换

通过学习到的对齐矩阵 $\boldsymbol{W}_l^*$，我们可以在推理过程中通过将源语言输入的表示转换到目标表示空间，来增强大语言模型对源语言输入的处理能力。

我们将测试输入 $\boldsymbol{q}^s$ 在源语言中，于大语言模型的第 l 层最后一个词元的隐藏状态表示为 $\boldsymbol{h}^{\mathrm{s}}_{q,l}$，然后使用对齐矩阵 $\boldsymbol{W}_l^*$ 将此源语言表示投影到目标表示空间：

$$\hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l} = \boldsymbol{W}_l^* \boldsymbol{h}^{\mathrm{s}}_{q,l} \tag{3}$$

为了在第 l 层使用干预向量 $\hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l}$ 执行跨语言干预，我们通过将其与目标语言中的投影隐藏状态 $\hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l}$ 混合，来调整源语言 $\boldsymbol{h}^{\mathrm{s}}_{q,l}$ 中的原始隐藏状态。此调整由一个超参数 α 控制，该参数平衡了源语言与目标语言隐藏状态之间的影响：

$$\boldsymbol{h}^{\mathrm{mix}}_{q,l} = \boldsymbol{h}^{\mathrm{s}}_{q,l} + \alpha \hat{\boldsymbol{h}}^{\mathrm{t}}_{q,l} \tag{4}$$

这里，方程4表示源语言表示向目标语言表示偏移了 α 倍。

最小干预解码在本工作中，我们仅对 $\boldsymbol{q}^{\mathrm{s}}$ 的最后一个词元进行一次干预，即在LLM的第 l层，将测试输入 $\boldsymbol{q}^s$ 中的$\boldsymbol{h}^{\mathrm{s}}_{q,l}$ 替换为$\boldsymbol{h}^{\mathrm{mix}}_{q\ l}$,。通过这种方式，我们可以有效干预模型输出，同时保留源语言中的特征。

与ITI和CAA的比较最近，ITI（Li等人, 2023b）和CAA（Rimsky等人, 2024）被提出作为干预方法，分别通过操纵选定的注意力头和隐藏状态来改变模型行为。INCLINE与ITI和CAA不同，主要有三点区别。首先，ITI和CAA使用学习到的**静态**干预向量来改变模型行为，而INCLINE利用一组对齐矩阵来动态地将输入表示从源语言对齐到目标语言。其次，ITI和CAA在整个指令后的所有词元位置应用干预向量，这可能在推理过程中造成过度扰动。相比之下，INCLINE仅对输入的最后一个词元执行一次干预。此外，与ITI和CAA仅针对有限数量的层不同，INCLINE修改所有层的表示。这些修改使大语言模型能够全面利用其目标语言能力进行多语言预测。

4 实验

在本节中，我们将介绍实验设置（第4.1节），并在第4.2节中展示结果。

4.1 实验设置

本节将介绍我们的评估任务、模型骨干、INCLINE的实现细节以及基线。

评估任务我们在九个不同的下游任务上进行了广泛的评估，这些任务分为两类：

- **判别性任务**：XCOPA(Ponti等人,2020)、XStoryCloze（Lin等人，2021b)、XWinograd（Lin等人, 2021b)、XCSQA（Lin等人, 2021a)、XNLI (Conneau等人, 2018)；
- **生成式任务**：MZsRE（Wang等人,2024b)、Flores（Goyal等人，2021年)、WMT23 (Kocmi等人, 2023)、MGSM(Shi等人,2022a)。

这些任务涵盖21种语言，包括英语(en)、阿拉伯语(ar)、德语(de)、希腊语(el)、西班牙语(es)、爱沙尼亚语(et)、法语(fr)、印地语(hi)、印尼语(id)、意大利语(it)、日语(ja)、荷兰语(nl)、葡萄牙语(pt)、俄语(ru)、斯瓦希里语(sw)、泰米尔语(ta)、泰语(th)、土耳其语(tr)、乌克兰语(uk)、越南语(vi)和中文(zh)。关于这些任务的更多细节，我们将在附录A中提供。

模型骨干 在本工作中，除非另有说明，我们主要使用BLOOMZ-7B1-MT作为所有基线方法的模型骨干。为了展示INCLINE在各种模型骨干上的有效性，我们纳入了四个额外的大语言模型：LLAMA3.1-8B-INSTRUCT（Dubey等人，2024）、LLAMA2-7B-CHAT（Touvron等人，2023）、MISTRAL-7B-INSTRUCT（Jiang等人，2023）、FALCON-7B-INSTRUCT（Almazrouei等人，2023）。我们在第6节中呈现这些结果。对于MGSM任务，我们采用MATHOCTOPUS（Chen等人，2023），[2]

[2]https://huggingface.co/Mathoctopus/Parallel_7B

	XCOPA			XStoryCloze			XWinograd			XCSQA			XNLI		
	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}
BASELINE	61.62	69.00	52.40	74.96	77.83	57.78	57.05	59.71	53.06	47.35	55.31	34.62	46.48	50.04	41.48
MT-GOOGLE	73.31†	73.52	73.05†	76.63	76.05	80.08†	57.63	57.12	57.90†	58.52†	54.84	64.40†	50.72	49.80	52.00
MT-LLM	59.84	67.16	50.70	79.41	82.23	62.48	43.02	41.67	45.04	30.73	35.38	23.30	43.64	47.83	37.77
Intervention Methods															
ITI	60.91	67.56	52.60	76.38	79.33	58.70	48.24	58.37	33.06	46.32	55.33	31.92	46.32	49.51	41.84
CAA	63.96	71.80	54.15	78.16	80.92	61.61	58.42	60.70	55.01	47.97	56.01	35.10	46.17	50.92	39.52
INCLINE	**65.22** **(+3.60)**	**72.56** **(+3.56)**	**56.05** **(+3.65)**	**79.92** **(+4.96)**	**82.03** **(+4.20)**	**67.24** **(+9.46)**	**59.35**† **(+2.30)**	**62.04**† **(+2.33)**	**55.32** **(+2.26)**	**48.45** **(+1.10)**	**56.45**† **(+1.14)**	**35.64** **(+1.02)**	**48.12** **(+1.64)**	**51.44** **(+1.40)**	**43.47** **(+1.99)**
SFT	66.89	76.84	54.45	87.36	89.50	74.52	43.78	48.63	36.50	42.18	47.95	32.96	69.68	76.76	59.76
SFT +INCLINE	**69.24** **(+2.35)**	**79.28**† **(+2.44)**	**61.22** **(+6.77)**	**88.11**† **(+0.75)**	**90.00**† **(+0.50)**	**76.77** **(+2.25)**	**49.84** **(+6.06)**	**57.58** **(+8.95)**	**38.24** **(+1.74)**	**42.55** **(+0.37)**	**48.38** **(+0.43)**	**33.22** **(+0.26)**	**71.17**† **(+1.49)**	**77.83**† **(+1.07)**	**61.84**† **(+2.08)**

Table 1: Main results of discriminative tasks. All the tasks are evaluated using accuracy. † denotes the best results. μ_{ALL}, μ_{SEEN}, and μ_{UNSEEN} indicate the macro-average of results across all the languages, the seen languages, and the unseen languages, respectively.

a specialized model fine-tuned from LLAMA2-7B for mathematical reasoning tasks, as the backbone.

INCLINE (Ours) In this work, we mainly focus on aligning the low-performing language (source) representations closer to the English (target) representations, as LLMs are predominantly English-centric. For training the alignment matrices between languages, we randomly sample 500 parallel sentence pairs for each language pair involving English and other languages. These pairs are sourced from the News Commentary v16 dataset (Barrault et al., 2019), and for languages not covered by this dataset, we use the CCAligned dataset (El-Kishky et al., 2020). Following Rimsky et al. (2024), the value of the α controlling the intervention strength is in the range from -1 to 1 and determined by the validation results for each language across tasks. We use one A100 GPU (40G) for all experiments.

Baselines We compare INCLINE against several established techniques on the zero-shot setting: (1) **BASELINE** indicates the predictions given by the original BLOOMZ-7B1-MT; (2) **MT-GOOGLE** utilizes GOOGLE TRANSLATE to translate non-English questions into English; (3) **MT-LLM** leverages BLOOMZ-7B1-MT to translate questions in non-English languages into English, employing the structured prompt template "`{Source Language}: {Inputs} English:`"; (4) **SFT** represents the task-specific supervised fine-tuning (SFT) involving updating all parameters of the LLM on the English training set for each downstream task individually with the hyperparameters described in Appendix B and evaluating the resulting model on the multilingual test sets; (5) **ITI** (Li et al., 2023b) is an intervention method that identifies attention heads with high linear probing accuracy for truthfulness and adjusts activations along these truth-correlated directions during inference. Originally used to shift models from generating false statements to truthful ones, we adapt it to encourage the generation of English text over non-English text. (6) **CAA** (Rimsky et al., 2024) employs the mean difference in hidden states between positive and negative examples from additional training data as an intervention vector to adjust the model's behavior towards the desired direction. Initially designed for monolingual alignment-relevant tasks, we utilize it to shift the model's output from non-English to English.

4.2 Results

In this section, we present our results on the discriminative tasks (Table 1) and generative tasks (Table 2). Furthermore, we also categorize the languages involved in the downstream tasks into two groups based on the training data of BLOOMZ-7B1-MT: *seen languages* (ar, es, fr, hi, id, pt, sw, ta, vi, and zh) and *unseen languages* (de, el, et, it, ja, nl, ru, th, tr, and uk). The breakdown results are provided in Table 7 (see Appendix C).

INCLINE significantly improves discriminative task performance. The experimental results in Table 1 clearly demonstrate the effectiveness of INCLINE. Although methods like SFT, MT-GOOGLE, and MT-LLM achieve high performance, they come with substantial costs, including the need for extensive fine-tuning of LLMs and reliance on third-party tools. Activation intervention methods, such as ITI and CAA, offer a more cost-effective solution but yield only minimal improvements, indicating a potential inadequacy in capturing the complexities of multilingual tasks. In contrast, INCLINE provides significant perfor-

	XCOPA			XStoryCloze			XWinograd			XCSQA			XNLI		
	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}
基线	61.62	69.00	52.40	74.96	77.83	57.78	57.05	59.71	53.06	47.35	55.31	34.62	46.48	50.04	41.48
MT-GOOGLE	73.31†	73.52	73.05†	76.63	76.05	80.08†	57.63	57.12	57.90†	58.52†	54.84	64.40†	50.72	49.80	52.00
MT-LLM	59.84	67.16	50.70	79.41	82.23	62.48	43.02	41.67	45.04	30.73	35.38	23.30	43.64	47.83	37.77
干预方法															
ITI	60.91	67.56	52.60	76.38	79.33	58.70	48.24	58.37	33.06	46.32	55.33	31.92	46.32	49.51	41.84
CAA	63.96	71.80	54.15	78.16	80.92	61.61	58.42	60.70	55.01	47.97	56.01	35.10	46.17	50.92	39.52
INCLINE	**65.22** **(+3.60)**	**72.56** **(+3.56)**	**56.05** **(+3.65)**	**79.92** **(+4.96)**	**82.03** **(+4.20)**	**67.24** **(+9.46)**	59.35† **(+2.30)**	62.04† **(+2.33)**	**55.32** **(+2.26)**	**48.45** **(+1.10)**	56.45† **(+1.14)**	**35.64** **(+1.02)**	**48.12** **(+1.64)**	**51.44** **(+1.40)**	**43.47** **(+1.99)**
SFT	66.89	76.84	54.45	87.36	89.50	74.52	43.78	48.63	36.50	42.18	47.95	32.96	69.68	76.76	59.76
监督微调 +INCLINE	**69.24** **(+2.35)**	79.28† **(+2.44)**	**61.22** **(+6.77)**	88.11† **(+0.75)**	90.00† **(+0.50)**	**76.77** **(+2.25)**	**49.84** **(+6.06)**	**57.58** **(+8.95)**	**38.24** **(+1.74)**	**42.55** **(+0.37)**	**48.38** **(+0.43)**	**33.22** **(+0.26)**	71.17† **(+1.49)**	77.83† **(+1.07)**	61.84† **(+2.08)**

表1：判别式任务的主要结果。所有任务均使用准确率进行评估。† 表示最佳结果。μ_{ALL}、μ_{SEEN}和 μ_{UNSEEN} 分别表示所有语言、已见语言和未见语言结果的宏平均。

一个专门为数学推理任务从LLAMA2-7B微调而来的模型，作为骨干。

INCLINE (本工作) 在本工作中，我们主要关注将低性能语言（源语言）的表示与英语（目标语言）的表示更紧密地对齐，因为大语言模型主要以英语为中心。为了训练语言间的对齐矩阵，我们为每个涉及英语和其他语言的语言对随机抽取500个平行句子对。这些句子对来源于News Commentary v16 数据集(Barrault等人, 2019年)，对于该数据集未覆盖的语言，我们使用CCAligned 数据集(El-Kishky等人,2020)。遵循Rimsky等人(2024)，控制干预强度 α 的值在-1到1的范围内，并由每个语言跨任务的验证结果确定。我们使用一块A100 GPU (40G) 进行所有实验。

基线 我们在零样本设置下将INCLINE与几种成熟技术进行比较：(1) **BASELINE** 表示原始BLOOMZ-7B1-MT给出的预测；(2)**MT-GOOGLE** 利用GOOGLE TRANSLATE 将非英语问题翻译成英语；(3)**MT-LLM** 利用BLOOMZ-7B1-MT 将非英语语言的问题翻译成英语，采用结构化提示模板"`{Source Language}: {Inputs} English:`"；(4)**SFT** 表示任务特定的监督微调（SFT），涉及使用附录B中描述的超参数，为每个下游任务单独更新大语言模型在英语训练集上的所有参数，并在多语言测试集上评估所得模型；(5) **ITI**(Li等人,2023b)是一种干预方法，它识别出具有高线性探测准确率以衡量真实性的注意力头，并在推理过程中沿这些与真实性相关的方向调整激活。该方法最初用于将模型从生成虚假陈述转向生成真实陈述，我们将其调整为鼓励生成英语文本而非非英语文本。(6) **CAA** (Rimsky等人, 2024)使用来自额外训练数据的正负示例之间隐藏状态的平均差异作为干预向量，以将模型的行为调整至期望方向。该方法最初为单语对齐相关任务设计，我们利用它将模型的输出从非英语转向英语。

4.2 结果

在本节中，我们展示了在判别式任务（表1）和生成式任务（表2）上的结果。此外，我们还根据BLOOMZ-7B1-MT的训练数据，将下游任务涉及的语言分为两组：已见语言（阿拉伯语、西班牙语、法语、印地语、印尼语、葡萄牙语、斯瓦希里语、泰米尔语、越南语和中文）和未见语言（德语、希腊语、爱沙尼亚语、意大利语、日语、荷兰语、俄语、泰语、土耳其语和乌克兰语）。详细结果见表7（参见附录C）。

INCLINE显著提升了判别性任务性能。实验结果显示，表1清晰地证明了INCLINE的有效性。尽管SFT、MT-GOOGLE和MT-LLM等方法取得了高性能，但它们伴随着巨大的成本，包括需要对大语言模型进行广泛的微调以及依赖第三方工具。激活干预方法，如ITI和CAA，提供了更具成本效益的解决方案，但仅带来微小的改进，这表明在捕捉多语言任务的复杂性方面可能存在不足。相比之下，INCLINE通过增强推理时间内的多语言表示对齐，提供了显著的性能提升，

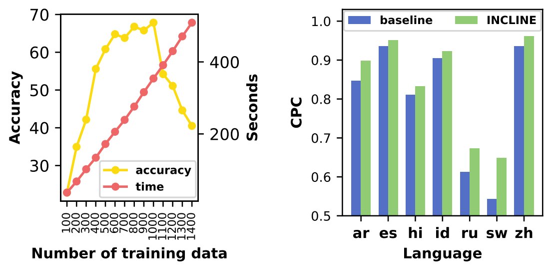
	MZsRE			Flores			WMT23			MGSM		
	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}
BASELINE	39.96	45.79	32.67	46.09	58.57	21.12	13.78	14.39	13.63	39.35	39.80	38.90
MT-GOOGLE	73.56†	72.76†	74.56†	-	-	-	-	-	-	46.70†	47.70†	45.70†
MT-LLM	33.18	39.25	25.61	-	-	-	-	-	-	21.40	30.00	12.80
Intervention Methods												
ITI	36.31	41.72	29.54	2.85	2.97	1.95	2.34	3.16	2.13	40.50	41.90	39.10
CAA	42.88	50.17	33.78	47.87	60.63	16.75	13.74	14.86	13.46	39.43	40.85	38.00
INCLINE	**43.22 (+3.26)**	**50.21 (+4.42)**	**34.49 (+1.82)**	**48.19† (+2.10)**	**61.28† (+2.71)**	**22.00† (+0.88)**	**14.23† (+0.45)**	**15.05† (+0.66)**	**14.02† (+0.39)**	**42.85 (+3.50)**	**43.30 (+3.50)**	**42.40 (+3.50)**

Table 2: Main results of generative tasks. † denotes the best results. μ_{ALL}, μ_{SEEN}, and μ_{UNSEEN} indicate the macro-average of results across all the languages, the seen languages, and the unseen languages, respectively. We use Exact Match (EM) to evaluate MZsRE, use BLEU to evaluate Flores and WMT23, and use accuracy to evaluate MGSM.

mance gains by enhancing multilingual representation alignment at inference time without requiring extensive resources or dependencies. This results in a more efficient improvement in multilingual performance. For example, INCLINE increases the average accuracy by +4.96 on XStoryCloze. Additionally, it delivers improvements of +4.20 and +9.46 for seen and unseen languages, respectively. Moreover, INCLINE can further improve the performance of the task-specific SFT.

INCLINE significantly enhances generative task performance. The experimental results presented in Table 2 suggest the effectiveness of INCLINE in enhancing performance across generative tasks. Unlike ITI and CAA, which show only marginal improvements similar to those observed in discriminative tasks, INCLINE appears to achieve substantial advancements. Notably, ITI seems to struggle significantly in machine translation tasks, such as Flores and WMT23, highlighting its limitations. Furthermore, INCLINE reportedly boosts accuracy in the MGSM task by up to +3.50 across various languages. This finding suggests that, although the mathematical capabilities are independent from the languages, understanding the questions written in different languages still requires language-specific knowledge. INCLINE successfully transfers the LLMs' natural language understanding capabilities from English to other languages. It is important to note that SFT is not evaluated on generative tasks because there are no training sets associated with these tasks.

In summary, these results demonstrate that INCLINE offers a significant improvement in both discriminative and generative tasks by effectively aligning multilingual representations.



(a) Training cost (b) Prediction consistency

Figure 3: (a) Training costs of INCLINE with regard to the number of parallel sentences and time used for training alignment matrices. INCLINE is evaluated on XStoryCloze in Swahili. (b) Correct Prediction Consistency (CPC) between non-English and English on XStoryCloze for the model using INCLINE.

5 Analysis

In this section, we conduct an in-depth analysis of INCLINE, focusing on four key aspects: computational costs, enhanced consistency after intervention, the impacts of the intervened components of LLMs, and the choice of intervention strength α. This analysis provides a comprehensive understanding of how INCLINE operates and its implications for model performance and efficiency.

INCLINE is highly efficient for training and introduces only marginal overhead for inference. To analyze the relationship between computational costs and accuracy, we measure both the training and inference costs of our method, INCLINE, using the XStoryCloze task in Swahili. As shown in Figure 3(a), increasing the amount of training data does not necessarily lead to improved accuracy, even though the training time is directly proportional to the number of samples. In our study, we empirically determine that using 500 samples for training the alignment matrices provides the best

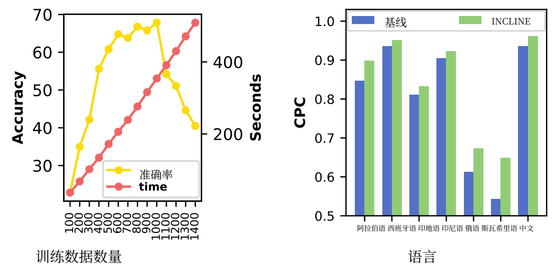
	MZsRE			Flores			WMT23			MGSM		
	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}	μ_{ALL}	μ_{SEEN}	μ_{UNSEEN}
基线	39.96	45.79	32.67	46.09	58.57	21.12	13.78	14.39	13.63	39.35	39.80	38.90
MT-GOOGLE	73.56†	72.76†	74.56†	-	-	-	-	-	-	46.70†	47.70†	45.70†
MT-LLM	33.18	39.25	25.61	-	-	-	-	-	-	21.40	30.00	12.80
干预方法												
ITI	36.31	41.72	29.54	2.85	2.97	1.95	2.34	3.16	2.13	40.50	41.90	39.10
CAA	42.88	50.17	33.78	47.87	60.63	16.75	13.74	14.86	13.46	39.43	40.85	38.00
INCLINE	**43.22 (+3.26)**	**50.21 (+4.42)**	**34.49 (+1.82)**	**48.19† (+2.10)**	**61.28† (+2.71)**	**22.00† (+0.88)**	**14.23† (+0.45)**	**15.05† (+0.66)**	**14.02† (+0.39)**	**42.85 (+3.50)**	**43.30 (+3.50)**	**42.40 (+3.50)**

表2：生成式任务的主要结果。† 表示最佳结果。μ_{ALL}、μ_{SEEN}和 μ_{UNSEEN} 分别表示所有语言、已见语言和未见语言结果的宏观平均值。我们使用精确匹配（EM）来评估MZsRE，使用BLEU来评估Flores和WMT23，并使用准确率来评估MGSM。

且无需大量资源或依赖。这带来了多语言性能的更高效提升。例如，INCLINE在XStoryCloze上将平均准确率提高了 +4.96。此外，它分别为已见语言和未见语言带来了 +4.20和 +9.46的改进。而且，INCLINE还能进一步提升任务特定SFT的性能。

INCLINE显著提升了生成式任务的性能。实验结果显示，表2表明INCLINE在提升生成式任务性能方面具有显著效果。与ITI和CAA仅在判别性任务中观察到类似的微小改进不同，INCLINE似乎取得了实质性进展。值得注意的是，ITI在机器翻译任务中似乎遇到显著困难，例如Flores 和WMT23，这突显了其局限性。此外，据报道，INCLINE将MGSM 任务的准确率提升了高达 +3.50，覆盖多种语言。这一发现表明，尽管数学能力与语言无关，但理解用不同语言书写的问题仍然需要语言特定的知识。INCLINE成功地将大语言模型（LLM）的自然语言理解能力从英语迁移到其他语言。需要注意的是，监督微调（SFT）未在生成式任务中进行评估，因为这些任务没有相关的训练集。

总之，这些结果表明，INCLINE通过有效对齐多语言表示，在判别式任务和生成式任务上都提供了显著改进。



(a) 训练成本 (b) 预测一致性

图3: (a) INCLINE的训练成本与平行句数量以及用于训练对齐矩阵的时间的关系。INCLINE在斯瓦希里语的XStoryCloze上进行评估。(b) 使用INCLINE的模型在XStoryCloze上非英语与英语之间的正确预测一致性(CPC)。

5 分析

在本节中，我们对INCLINE进行深入分析，重点关注四个关键方面：计算成本、干预后的一致性增强、大语言模型干预组件的影响，以及干预强度 α的选择。此分析全面理解INCLINE如何运作及其对模型性能和效率的影响。

INCLINE在训练方面非常高效，并且仅引入微小的推理开销。为了分析计算成本与准确率之间的关系，我们使用XStoryCloze 任务在斯瓦希里语中测量了我们的方法INCLINE的训练和推理成本。如图3(a)所示，增加训练数据的量并不一定会提高准确率，尽管训练时间与样本数量直接成正比。在我们的研究中，我们通过实证确定，使用500个样本来训练对齐矩阵能提供最佳的

	XCOPA	XCSQA	Flores	MGSM
BASELINE	61.60	47.35	46.09	39.35
INCLINE				
├ INCLINE-HIDDEN	**65.22**	**48.45**	**48.19**	**42.85**
├ INCLINE-ATTN	63.87	48.18	47.54	41.55
├ INCLINE-FFN	64.20	47.96	46.10	41.80
└ INCLINE-EMB	63.16	47.59	39.23	40.90

Table 3: The averaged results of XStoryCloze, XCSQA, Flores, MGSM tasks with four configurations for INCLINE given by BLOOMZ-7B1-MT.

balance between performance gains and computational costs. Consequently, the training process takes only 172 seconds. During inference, our approach involves a single intervention at the last token, resulting in a time complexity of $O(1)$. This method incurs only a 12% increase in inference time, taking 0.80 seconds per item compared to 0.71 seconds without it, thereby maintaining a low inference cost. More results are provided in Appendix G.

INCLINE effectively enhances the consistency of correct predictions between non-English languages (source) and English (target). Recent non-English test sets are commonly translated from their English versions, either by humans or machines, creating parallel datasets. To quantify the alignment between non-English languages (source) and English (target), we propose using the Correct Prediction Consistency (CPC) rate. This metric measures the proportion of questions correctly answered in both languages, with a higher CPC rate indicating better alignment. The results in Figure 3(b) demonstrate that CPC significantly improves after intervention by INCLINE, suggesting that INCLINE effectively aligns non-English representations with English ones for more accurate predictions. Notably, CPC for Swahili (sw) increases from 0.54 to 0.65 with INCLINE, showing its effectiveness for low-resource languages.

Intervening on hidden states yields the greatest performance improvements. We apply INCLINE to various components of LLMs, including the hidden states (INCLINE-HIDDEN), the outputs of attention heads (INCLINE-ATTN), the outputs of FFN blocks (INCLINE-FFN), and the embeddings (INCLINE-EMB). The results presented in Table 3 indicate that intervening on the hidden states (INCLINE-HIDDEN) leads to the most significant improvements across multilingual tasks. This finding suggests that hidden states can capture

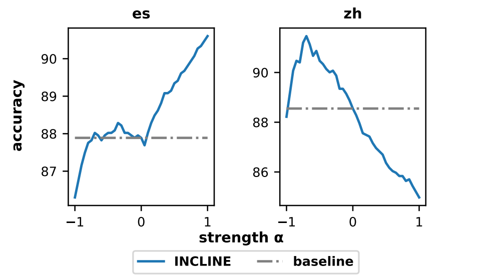


Figure 4: The accuracy changed with hyperparameter α on the XStoryCloze task with BLOOMZ-7B1-MT.

	ar	es	hi	id	ru	sw	zh	AVG
				BLOOMZ-7B1-MT				
BASELINE	79.22	87.89	76.37	84.45	57.78	50.50	88.55	74.96
INCLINE	**83.12**	**90.60**	**81.47**	**86.10**	**67.24**	**59.70**	**91.20**	**79.92**
				LLAMA3.1-8B-INSTRUCT				
BASELINE	86.50	91.73	84.84	37.46	66.98	54.00	92.39	73.41
INCLINE	**87.36**	**92.39**	**85.31**	**64.53**	**73.73**	**55.66**	**92.72**	**78.81**
				LLAMA2-7B-CHAT				
BASELINE	49.37	47.25	39.25	48.18	34.94	0.93	55.53	39.35
INCLINE	**51.42**	**56.65**	**47.25**	**49.97**	**41.03**	**17.67**	**60.69**	**46.38**
				MISTRAL-7B-INSTRUCT				
BASELINE	18.33	81.34	24.95	76.64	83.65	2.58	90.07	53.94
INCLINE	**36.71**	**84.23**	**35.77**	**80.18**	**85.13**	**25.71**	**90.34**	**62.58**
				FALCON-7B-INSTRUCT				
BASELINE	53.61	58.31	53.21	55.59	54.60	51.16	54.00	54.35
INCLINE	**54.33**	**61.81**	**54.33**	**58.04**	**57.91**	**53.47**	**59.70**	**57.09**

Table 4: The results of XStoryCloze dataset with five LLM backbones.

comprehensive semantic information that is crucial for cross-lingual alignment. While INCLINE-ATTN, INCLINE-FFN, and INCLINE-EMB also enhance performance, their performance gains vary across different tasks. These findings justify our design choice of using hidden states in INCLINE.

The value of α varies across languages and depends on language relatedness. In this study, we introduce α to control the strength of intervention in Equation 4. To investigate the impact of α, we conduct a grid search to find the optimal α values across the languages in XStoryCloze. We present the results for Spanish and Chinese in Figure 4. We observe that the optimal α values for these two languages are opposite: positive for Spanish and negative for Chinese. These findings suggest that the value of α is likely to depend on language relatedness, as both Spanish and English belong to the Indo-European language family, while Chinese belongs to the Sino-Tibetan language family. Results for more languages are provided in Appendix D.

6 Discussions

In this section, we conduct a series of experiments to investigate how variations in LLMs, model sizes,

	XCOPA	XCSQA	Flores	MGSM
基线	61.60	47.35	46.09	39.35
INCLINE				
├ INCLINE-HIDDEN	**65.22**	**48.45**	**48.19**	**42.85**
├ INCLINE-ATTN	63.87	48.18	47.54	41.55
├ INCLINE-FFN	64.20	47.96	46.10	41.80
└ INCLINE-EMB	63.16	47.59	39.23	40.90

表3：BLOOMZ-7B1-MT给出的INCLINE四种配置在XStoryCloze、XCSQA、Flores、MGSM任务上的平均结果。

性能提升与计算成本之间的平衡。因此，训练过程仅需172秒。在推理过程中，我们的方法涉及在最后一个词元处进行单次干预，导致时间复杂度为 $O(1)$。这种方法仅使推理时间增加12%，每个项目耗时0.80秒，相比之下，不使用该方法时为0.71秒，从而保持了较低的推理成本。更多结果见附录G。

INCLINE有效增强了非英语语言（源语言）与英语（目标语言）之间正确预测的一致性。 近期的非英语测试集通常是从其英语版本翻译而来，无论是通过人工还是机器翻译，从而创建了平行数据集。为了量化非英语语言（源语言）与英语（目标语言）之间的对齐程度，我们提出了使用正确预测一致性（CPC）率。该指标衡量了在两种语言中均被正确回答的问题比例，CPC率越高表示对齐效果越好。图3(b)中的结果表明，经过INCLINE干预后，CPC显著提升，这表明INCLINE有效地将非英语表示与英语表示对齐，以实现更准确的预测。值得注意的是，斯瓦希里语（sw）的CPC在使用INCLINE后从0.54提升至0.65，显示了其对低资源语言的有效性。

对隐藏状态进行干预能带来最大的性能提升。 我们将INCLINE应用于大语言模型的各个组件，包括隐藏状态（INCLINE-HIDDEN）、注意力头的输出（INCLINE-ATTN）、前馈网络块的输出（INCLINE-FFN）以及嵌入（INCLINE-EMB）。表3中呈现的结果表明，对隐藏状态（INCLINE-HIDDEN）进行干预能在多语言任务中带来最显著的改进。这一发现表明隐藏状态能够捕捉

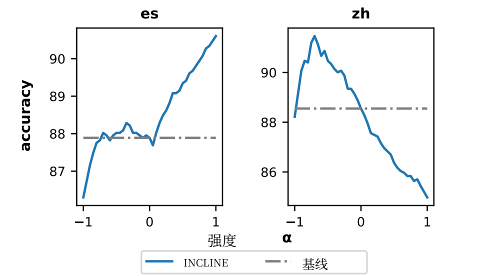


图4：在XStoryCloze任务上，使用BLOOMZ-7B1-MT时，准确率随超参数 α的变化情况。

	ar	es	hi	id	ru	sw	zh	AVG
				BLOOMZ-7B1-MT				
B基线	79.22	87.89	76.37	84.45	57.78	50.50	88.55	74.96
INCLINE	**83.12**	**90.60**	**81.47**	**86.10**	**67.24**	**59.70**	**91.20**	**79.92**
				LLAMA3.1-8B-INSTRUCT				
B基线	86.50	91.73	84.84	37.46	66.98	54.00	92.39	73.41
INCLINE	**87.36**	**92.39**	**85.31**	**64.53**	**73.73**	**55.66**	**92.72**	**78.81**
				LLAMA2-7B-CHAT				
基线	49.37	47.25	39.25	48.18	34.94	0.93	55.53	39.35
INCLINE	**51.42**	**56.65**	**47.25**	**49.97**	**41.03**	**17.67**	**60.69**	**46.38**
				MISTRAL-7B-INSTRUCT				
B基线	18.33	81.34	24.95	76.64	83.65	2.58	90.07	53.94
INCLINE	**36.71**	**84.23**	**35.77**	**80.18**	**85.13**	**25.71**	**90.34**	**62.58**
				FALCON-7B-INSTRUCT				
基线	53.61	58.31	53.21	55.59	54.60	51.16	54.00	54.35
INCLINE	**54.33**	**61.81**	**54.33**	**58.04**	**57.91**	**53.47**	**59.70**	**57.09**

表4：XStoryCloze数据集在五个大语言模型骨干上的结果。

全面的语义信息，这对于跨语言对齐至关重要。虽然INCLINE-ATTN、INCLINE-FFN和INCLINE-EMB也能提升性能，但它们的性能增益在不同任务中有所差异。这些发现证明了我们在INCLINE中使用隐藏状态的设计选择是合理的。

该值 α 因语言而异，并取决于语言间的亲缘关系。 在本研究中，我们引入 α 来控制方程4中干预的强度。为了探究 α的影响，我们进行了网格搜索，以找到在XStoryCloze各语言中的最优 α 值。我们在图4中展示了西班牙语和中文的结果。我们观察到，这两种语言的最优 α 值相反：西班牙语为正值，中文为负值。这些发现表明， α 的值很可能取决于语言亲缘关系，因为西班牙语和英语同属印欧语系，而中文则属于汉藏语系。更多语言的结果见附录D。

6 讨论

在本节中，我们进行了一系列实验，以研究大语言模型、模型规模、

	ar	el	es	fr	hi	ru	tr	vi	zh	AVG
BASELINE	66.59	15.30	48.52	67.86	71.97	35.66	12.38	40.40	56.11	46.09
INCLINE	68.68	15.63	50.79	69.93	76.92	37.95	12.42	43.11	58.27	48.19
INCLINE-FDEV	**73.95**	**15.76**	**56.11**	**75.84**	**77.85**	**39.33**	**12.92**	**46.49**	**60.19**	**50.94**

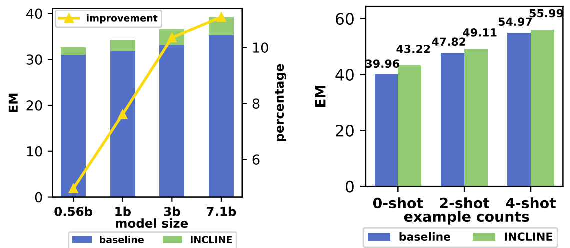
Table 5: The BLEU results of `Flores` dataset with INCLINE and INCLINE-FDEV.

in-context learning, and the data used for training alignment matrices affect our results. Additionally, we also explore using French as the target language (Appendix E).

INCLINE consistently enhances performance across multiple LLMs. To demonstrate the versatility of INCLINE across different LLMs, we apply it to another four high-performing models on the `XStoryCloze` task. As shown in Table 4, INCLINE consistently enhances performance compared to the BASELINE. Specifically, we observe increases of +4.96 for BLOOMZ-7B1-MT, +5.40 for LLAMA3.1-8B-INSTRUCT, +7.03 for LLAMA2-7B-CHAT, +8.64 for MISTRAL-7B-INSTRUCT, and +2.74 for FALCON-7B-INSTRUCT.

Larger LLMs benefit more from INCLINE. Building on the work of Wang et al. (2024b), who demonstrates a scaling relationship between the size of backbone models and their performance, we evaluate the impact of different model sizes within the BLOOMZ series on the `MZsRE` dataset. Our findings, illustrated in Figure 5(a), show that the relative performance gain of INCLINE over the baseline increases with the size of the backbone model. Specifically, the Exact Match (EM) scores (in the stacked columns) and the improvement percentages (in the line chart) indicate that larger models in the BLOOMZ series exhibit more significant enhancements when INCLINE is applied. This observation demonstrates that larger LLMs can benefit more from INCLINE.

INCLINE can further enhance model performance when combined with in-context learning. In-context learning (ICL) has been shown to improve the performance of LLMs on the `MZsRE` task (Wang et al., 2024b). Building upon this finding, we evaluate the effectiveness of combining INCLINE with ICL. As illustrated in Figure 5(b), INCLINE demonstrates enhanced performance, achieving an additional increase of +1.02 in average Exact Match (EM) score with four in-context examples compared to the baseline using ICL alone. While this improvement is smaller than the +3.26



(a) Various model sizes (b) In-context learning

Figure 5: (a) Exact Match (left y-axis) and relative improvements over the baseline (right y-axis) on `MZsRE` with respect to various model sizes of BLOOMZ. (b) Exact Match score for `MZsRE` dataset with INCLINE based on the zero-shot setting and few-shot settings given by BLOOMZ-7B1-MT.

increase observed in the zero-shot setting, it suggests that the benefits of INCLINE and ICL are complementary, with both methods capturing features from different perspectives. This highlights the versatility of INCLINE in various applications.

High-quality parallel sentences improve alignment in INCLINE. We explore how the quality of parallel sentences affects the performance of INCLINE. By default, the alignment matrices of INCLINE are trained using 500 random samples from the News Commentary dataset. To assess the impact of sentence quality, we also train the alignment matrices using 500 high-quality parallel sentences from the development set of `Flores`, which are carefully translated by professional human translators. We refer to this variant as INCLINE-FDEV. In Table 5, INCLINE-FDEV significantly outperforms both the standard INCLINE and BASELINE, highlighting the importance of high-quality parallel sentences.

7 Conclusion

In this paper, we introduce **Inference-Time Cross-Lingual Intervention (INCLINE)**, an innovative framework that bridges the performance gaps between high-performing and low-performing languages in LLMs. By training alignment matrices to

	ar	el	es	fr	hi	ru	tr	vi	zh	AVG
基线	66.59	15.30	48.52	67.86	71.97	35.66	12.38	40.40	56.11	46.09
INCLINE	68.68	15.63	50.79	69.93	76.92	37.95	12.42	43.11	58.27	48.19
INCLINE-FDEV	**73.95**	**15.76**	**56.11**	**75.84**	**77.85**	**39.33**	**12.92**	**46.49**	**60.19**	**50.94**

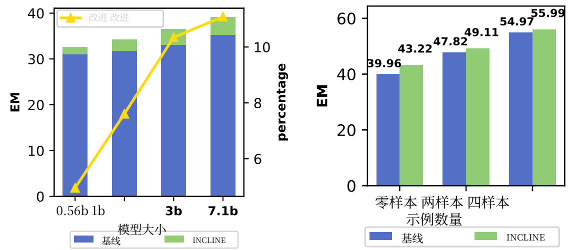
表5：Flores数据集上INCLINE与INCLINE-FDEV的BLEU结果。

上下文学习以及用于训练对齐矩阵的数据如何影响我们的结果。此外，我们还探讨了使用法语作为目标语言（附录E）。

INCLINE在多个大语言模型中持续提升性能。为展示INCLINE在不同大语言模型间的通用性，我们将其应用于XStoryCloze任务上的另外四个高性能模型。如表4所示，与基线相比，INCLINE持续提升了性能。具体而言，我们观察到BLOOMZ-7B1-MT提升了 +4.96，LLAMA3.1-8B-INSTRUCT提升了 +5.40，LLAMA2-7B-CHAT提升了 +7.03，MISTRAL-7B-INSTRUCT提升了 +8.64，以及FALCON-7B-INSTRUCT提升了+2.74。

更大的大语言模型从INCLINE中获益更多。基于Wang等人(2024b)的研究，他们展示了骨干模型大小与其性能之间的缩放关系，我们评估了BLOOMZ系列中不同模型大小对`MZsRE`数据集的影响。我们的发现，如图5(a)所示，表明INCLINE相对于基线的相对性能增益随着骨干模型大小的增加而增加。具体来说，精确匹配（EM）分数（在堆叠柱状图中）和改进百分比（在线图中）表明，BLOOMZ系列中更大的模型在应用INCLINE时表现出更显著的增强。这一观察结果表明，更大的大语言模型可以从INCLINE中获益更多。

INCLINE与上下文学习结合使用时，能进一步提升模型性能。上下文学习（ICL）已被证明能提升大语言模型在`MZsRE`任务上的表现（Wang等人，2024b）。基于这一发现，我们评估了将INCLINE与ICL结合使用的效果。如图5(b)所示，INCLINE展现出增强的性能，在使用四个上下文示例时，相比仅使用ICL的基线，平均精确匹配（EM）分数额外提升了 +1.02。虽然这一提升幅度小于 +3.26



(a) 各种模型大小 (b) 上下文学习

图5：(a) 在MZsRE数据集上，针对BLOOMZ的不同模型大小，精确匹配（左y轴）以及相对于基线的相对改进（右y轴）。(b) 基于BLOOMZ-7B1-MT提供的零样本设置和少样本设置，MZsRE数据集上使用INCLINE的精确匹配分数。

在零样本设置中观察到的增幅，但这表明INCLINE与ICL的益处是互补的，两种方法从不同角度捕捉特征。这突显了INCLINE在各种应用中的多功能性。

高质量的平行句能改善INCLINE中的对齐。我们探讨了平行句的质量如何影响INCLINE的性能。默认情况下，INCLINE的对齐矩阵是使用News Commentary数据集中的500个随机样本进行训练的。为了评估句子质量的影响，我们还使用了Flores开发集中由专业人工翻译人员精心翻译的500个高质量平行句来训练对齐矩阵。我们将此变体称为INCLINE-FDEV。在表5中，INCLINE-FDEV显著优于标准INCLINE和基线，突显了高质量平行句的重要性。

7 结论

本文介绍**推理时跨语言干预（INCLINE）**，这是一个创新的框架，旨在弥合大语言模型中高性能语言与低性能语言之间的性能差距。通过训练对齐矩阵，

transform source low-performing language representations into the target high-performing language representation space, INCLINE enhances performance on underrepresented languages without requiring additional training or fine-tuning of LLMs. Extensive experiments across nine benchmarks and five LLMs demonstrate that, INCLINE delivers significant improvements by up to +4.96 in terms of accuracy compared to strong baselines, while it only requires minimal computational costs.

8 Limitations

While INCLINE demonstrates significant enhancement for the multilingual tasks with cross-lingual intervention, the alignment matrices are trained for specific pairs of source and target languages. Future work will focus on developing multilingual alignment matrices that can accommodate multiple languages simultaneously, reducing the need for language pair-specific training and enhancing scalability. Implementing INCLINE requires access to the internal layers and representations of LLMs. For proprietary or closed-source models, or models accessible only through APIs without exposure of internal representations (e.g., GPT-4o), applying this method may not be feasible. How to perform cross-lingual alignment as a plug-and-play tool for all LLMs, including those with restricted access, requires further investigation.

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将源低性能语言表示转换为目标高性能语言表示空间，INCLINE提升了在代表性不足语言上的性能，而无需对大语言模型进行额外的训练或微调。在九个基准测试和五个大语言模型上进行的大量实验表明，与强大的基线相比，INCLINE在准确率方面带来了高达 +4.96的显著提升，同时仅需极少的计算成本。

8 局限性

虽然INCLINE通过跨语言干预在多语言任务上展现出显著的增强效果，但其对齐矩阵是针对特定的源语言和目标语言对进行训练的。未来的工作将侧重于开发能够同时适应多种语言的多语言对齐矩阵，从而减少对语言对特定训练的需求，并提升可扩展性。实施INCLINE需要访问大语言模型的内部层和表示。对于专有或闭源模型，或者仅能通过API访问且不暴露内部表示的模型（例如GPT-4o），应用此方法可能不可行。如何将跨语言对齐作为一种适用于所有大语言模型（包括那些访问受限的模型）的即插即用工具，需要进一步研究。

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A Details of Datasets

The tasks and the corresponding output format, prompt template, evaluation metrics, the number of languages are shown in Table 6.

B Hyperparameters for SFT

We fine-tune all parameters of LLMs using the AdamW optimizer with a learning rate of 2×10^{-6} and a batch size of 4. This process is conducted over three epochs on four NVIDIA A100 GPUs (80GB). During training, we use a linear learning rate schedule with a warm-up phase that constitutes 10% of the total training steps.

C Detailed Results of Intervention

The detailed results of BASELINE, MT-GOOGLE, MT-LLM, SFT, ITI, CAA, INCLINE and SFT +INCLINE for each languages across discriminative and generative tasks are shown in Table 7.

D The value of α across languages

We explore the optimal value of α for each language in `XStoryCloze` using grid search, as shown in Figure 6.

E Projection to Non-English

We have demonstrated the effectiveness of INCLINE in aligning representations from non-English to English. To further prove the generalizability of INCLINE with another high-performing language, we conduct an ablation study aligning representations of various languages with French. As shown in Table 8, INCLINE enhances translation performance to non-English languages, with an average BLEU score increase of +5.35. This further demonstrates that INCLINE can effectively align representations across different languages.

F Details of Visualizing

Following Li et al. (2023b), we use Linear Regression to examine multilingual input representations. For each English and corresponding Portuguese sample from the News Commentary dataset (a total of 500 items), we extract the hidden states at the last token to create a probing dataset for each layer. We randomly divide this dataset into training and validation sets in a 4:1 ratio and fit a binary linear classifier to the training set. Similar to principal component analysis (PCA), we train a second linear probe on the same dataset, constrained to be orthogonal to the first probe. This orthogonality ensures that the two probes capture distinct aspects of the data. Finally, we project the hidden states of each sample in the `MZsRE` test set onto the directions defined by the probes from the last layer, allowing us to visualize and analyze the multilingual representations effectively.

G Supplementary Results of Training Data Volume

We find that increasing the amount of training data for learning alignment matrices can unexpectedly degrade performance in Figure 3(a). To investigate this observation, we examine the learned alignment matrices from the last layer, using varying numbers of parallel sentences. Interestingly, as shown in Table 9, the absolute values of the learned alignment weights consistently increase as the number of parallel sentences increases. It is well known that large weights provide the capacity for the network to fit the training data closely. As a result, the learned alignment matrices tend to fit more closely to the distribution of the parallel sentences. Consequently, the distribution shift between the parallel sentences and downstream tasks is enlarged as the number of parallel sentences grows.

Data	200	500	700	1000	1200	1400
max	0.57	0.84	1.01	1.31	1.43	1.57
min	-0.54	-0.73	-0.88	-1.29	-1.49	-1.71
avg	2.9e-7	1.0e-6	1.4e-6	1.9e-6	2.9e-6	3.6e-6
std	0.005	0.009	0.011	0.014	0.016	0.019

Table 9: The learned alignment matrices from the last layer with using varying numbers of parallel sentences.

A 数据集详情

任务及相应的输出格式、提示模板、评估指标、语言数量如表6所示。

B 监督微调的超参数

我们使用AdamW优化器，学习率为 2×10^{-6}，批次大小为4，对大语言模型的所有参数进行微调。此过程在四块NVIDIA A100 GPU（80GB）上进行三个训练轮次。训练期间，我们采用线性学习率调度，其中预热阶段占总训练步数的10%。

C 干预的详细结果

B的详细结果ASELINE、MT-GOOGLE、MT-LLM、监督微调、ITI、CAA、INCLINE和监督微调+ INCLINE在每种语言上，针对判别式和生成式任务的结果显示于表7。

D α 在不同语言中的值

我们通过网格搜索探索了 α 在XStoryCloze中每种语言的最优值`XStoryCloze`，如图6所示。

E 向非英语语言的投影

我们已经证明了INCLINE在将非英语语言的表示与英语对齐方面的有效性。为了进一步证明INCLINE与另一种高性能语言的泛化能力，我们进行了一项消融研究，将各种语言的表示与法语对齐。如表8所示，INCLINE增强了对非英语语言的翻译性能，平均BLEU分数提高了 +50.35。这进一步证明INCLINE能够有效地在不同语言间对齐表示。

F 可视化细节

遵循Li等人(2023b)的方法，我们使用线性回归来检查多语言输入表示。对于来自News Commentary数据集的每个英语及对应葡萄牙语样本（共500项），我们提取最后一个词元的隐藏状态，为每一层创建一个探测数据集。我们将此数据集按4:1的比例随机划分为训练集和验证集，并在训练集上拟合一个二元线性分类器。类似于主成分分析（PCA），我们在同一数据集上训练第二个线性探针，约束其与第一个探针正交。这种正交性确保两个探针捕捉数据的不同方面。最后，我们将MZsRE测试集中每个样本的隐藏状态投影到由最后一层探针定义的方向上，从而能够有效地可视化和分析多语言表示。

G 训练数据量补充结果

我们发现，增加用于学习对齐矩阵的训练数据量可能会意外地降低图3(a)中的性能。为了研究这一观察结果，我们检查了使用不同数量平行句从最后一层学习到的对齐矩阵。有趣的是，如表9所示，学习到的对齐权重的绝对值随着平行句数量的增加而持续增加。众所周知，较大的权重为网络提供了紧密拟合训练数据的能力。因此，学习到的对齐矩阵往往更紧密地拟合平行句的分布。结果，随着平行句数量的增加，平行句与下游任务之间的分布偏移被放大了。

Data	200	500	700	1000	1200	1400
max	0.57	0.84	1.01	1.31	1.43	1.57
min	-0.54	-0.73	-0.88	-1.29	-1.49	-1.71
avg	2.9e-7	1.0e-6	1.4e-6	1.9e-6	2.9e-6	3.6e-6
std	0.005	0.009	0.011	0.014	0.016	0.019

表9：使用不同数量平行句从最后一层学习到的对齐矩阵。

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Dataset	Output	prompt	Metric	|L|
XCOPA	2-way class	Here is a premise: "{premise}". A: "{choice1}" B: "{choice2}" What is the {question}? "A" or "B"?	acc.	10
XStoryCloze	2-way class	{input} What is a possible continuation for the story given the following options? A: {quiz1} B: {quiz2}'	acc.	8
XWinograd	2-way class	{input} Replace the _ in the above sentence with the correct option: - {option1} - {option2}	acc.	6
XNLI	3-way class	Take the following as truth: {premise} Then the following statement: "{hypothesis}" is "true", "false", or "inconclusive"?	acc.	13
XCSQA	multi-choice	Question: {question} {choice} Answer:	acc.	14
MZsRE	answer	{context} Quesion: {question} Answer:	EM	10
Flores	answer	Translate the following sentence from {language} to English: {input}	BLEU	10
WMT23	answer	Translate the following sentence from {language} to English: {input}	BLEU	5
MGSM	answer	Write a response that appropriately completes the request in {language}. Please answer in {language}. ### Instruction: {query}### Response:	EM	9

Table 6: The nine datasets used to evaluate multilingual intervention. |L| indicates the number of languages. EM is the Exact Match score and acc. represents the accuracy.

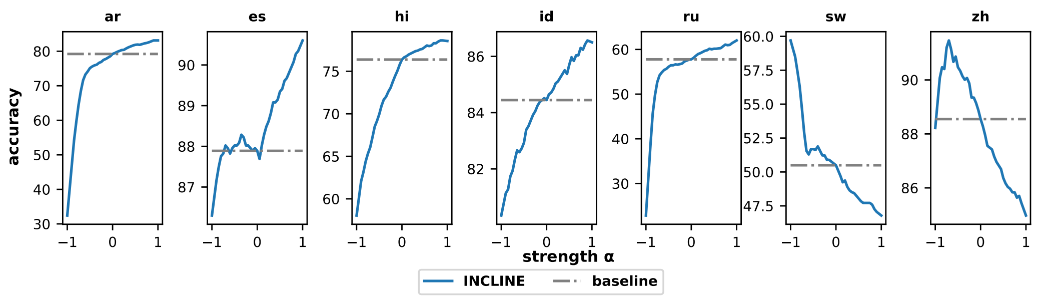

Figure 6: The accuracy changed with hyperparameter α on the XStoryCloze task.

数据集	输出	提示	指标	|L|
XCOPA	二分类	这里有一个前提："{premise}"。A: "{choice1}" B: "{choice2}" {question}是什么？"A"还是"B"?	acc.	10
XStoryCloze	二分类	{input} 根据以下选项，这个故事可能的后续发展是什么？A: {quiz1} B: {quiz2}	acc.	8
XWinograd	二分类	{input} 将上述句子中的_ 替换为正确的选项：- {option1} - {option2}	acc.	6
XNLI	三分类	将以下内容视为事实：{premise} 那么以下陈述："{hypothesis}" 是 "真"、"假" 还是 "不确定"?	acc.	13
XCSQA	多项选择	问题：{question} {choice} 答案：	acc.	14
MZsRE	答案	{context} 问题：{question} 答案：	EM	10
Flores	答案	将以下句子从{language}翻译成英语：{input}	BLEU	10
WMT23	答案	将以下句子从{language}翻译成英语：{input}	BLEU	5
MGSM	答案	请用{language}撰写一个恰当完成该请求的回复。请用{language}回答。### 指令：{query}### 响应：	EM	9

表6：用于评估多语言干预的九个数据集。 |L| 表示语言数量。EM是精确匹配分数，准确率代表准确率。

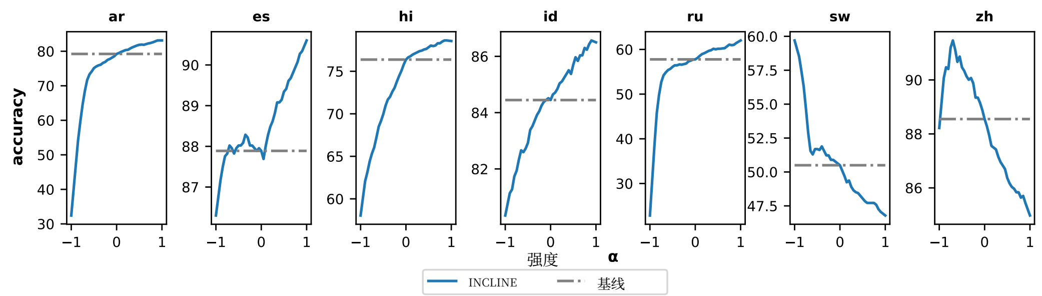

图6：在XStoryCloze任务上，准确率随超参数 α 的变化情况。

Discriminative tasks															
XCOPA	en	et	id	it	sw	ta	th	tr	vi	zh					AVG
Baseline	76.40	50.80	69.60	58.60	55.20	71.60	50.60	49.60	71.20	77.40					61.62
MT-Google	-	75.40†	75.00	76.00†	76.20†	62.20	62.40†	78.40†	76.40	77.80					73.31†
MT-LLM	-	44.80	69.80	59.40	60.20	71.20	47.40	51.20	61.60	73.00					59.84
SFT	86.40	50.60	78.40	67.80	59.00	77.20	47.60	53.00	83.00	84.60					66.80
ITI	-	50.80	70.80	60.00	55.40	63.20	49.00	50.60	69.00	79.40					60.91
CAA	-	51.20	72.20	61.20	59.20	73.00	52.20	52.00	74.80	79.80					63.96
INCLINE	-	**55.40**	**73.40**	**62.80**	**59.80**	**73.40**	**52.60**	**53.40**	**76.20**	**80.00**					**65.22**
SFT +INCLINE	-	**53.20**	**81.20†**	**65.80**	**60.80**	**85.00†**	**54.40**	**53.40**	**84.40†**	**85.00†**					**69.24**
XStoryCloze	en	ar	es	hi	id	ru	sw	zh							AVG
Baseline	91.46	79.22	87.89	76.37	84.45	57.78	50.50	88.55							74.96
MT-Google	-	79.48	81.34	50.69	80.81	80.08†	77.04	86.96							76.63
MT-LLM	-	81.80	86.83	82.59	83.59	62.48	73.66	84.91							79.41
SFT	94.11	90.47	92.85	88.22	91.59	74.52	81.14	92.72							87.36
ITI	-	78.23	90.54	80.28	85.70	58.70	52.55	88.68							76.38
CAA	-	86.04	90.47	79.15	88.22	61.61	52.61	89.01							78.16
INCLINE	-	**83.12**	**90.60**	**81.47**	**86.10**	**67.24**	**59.70**	**91.20**							**79.92**
SFT +INCLINE	-	**90.93†**	**92.98†**	**89.08†**	**91.99†**	**76.77**	**81.93†**	**93.05†**							**88.11†**
XWinograd	en	fr	ja	pt	ru	zh									AVG
Baseline	73.76	59.04	51.51	57.80	54.60	62.30									57.05
MT-Google	-	61.45	58.39†	59.32†	57.41	50.60									57.63
MT-LLM	-	54.22	47.86	33.08	42.22	37.70									43.02
SFT	78.06	62.65	14.91	43.35	58.09	39.89									43.78
ITI	-	54.22	51.51	57.79	14.60	63.10									48.24
CAA	-	60.24	52.87	58.17	57.14	63.69									58.42
INCLINE	-	**63.86†**	**53.18**	**58.56**	**57.46**	**63.69†**									**59.35†**
SFT +INCLINE	-	**63.86†**	**16.48**	**46.39**	**60.00†**	**62.50**									**49.84**
XCSQA	en	ar	de	es	fr	hi	it	ja	nl	pt	ru	sw	vi	zh	AVG
Baseline	76.50	52.40	33.90	64.30	63.30	48.50	41.30	36.00	28.70	61.30	33.20	40.50	55.20	57.00	47.35
MT-Google	-	61.60†	65.00†	68.00†	67.20†	32.10	68.70†	57.30†	66.50†	66.90†	64.50†	19.60	62.90†	60.40†	58.52†
MT-LLM	-	32.30	26.30	42.70	42.30	30.40	25.60	25.60	17.40	39.90	21.60	24.00	31.60	39.80	30.73
SFT	65.70	48.20	32.90	54.10	53.60	43.10	40.40	32.60	29.00	53.60	29.90	31.80	48.40	50.80	42.18
ITI	-	52.10	34.20	64.50	63.70	48.10	40.00	25.90	26.00	61.20	33.50	40.90	54.90	57.20	46.32
CAA	-	52.80	34.10	64.50	63.30	48.40	42.20	36.40	29.30	62.80	33.50	41.90	56.00	58.40	47.97
INCLINE	-	**53.20**	**34.90**	**65.00**	**63.80**	**48.80†**	**42.90**	**36.80**	**29.80**	**62.60**	**33.80**	**42.20†**	**57.30**	**58.70**	**48.45**
SFT +INCLINE	-	**48.50**	**33.30**	**54.40**	**53.70**	**43.90**	**40.60**	**33.00**	**29.30**	**53.70**	**29.90**	**32.50**	**49.10**	**51.20**	**42.55**
XNLI	en	ar	de	el	es	fr	hi	ru	sw	th	tr	vi	zh		AVG
Baseline	54.81	53.63	43.33	41.04	51.36	50.54	50.16	47.80	45.01	40.32	34.93	49.68	49.92		46.48
MT-Google	-	51.46	53.13	52.71	51.84	50.82	41.58	51.68	50.54	50.50	52.00	51.94	50.42		50.72
MT-LLM	-	46.87	43.25	36.29	52.12	51.40	45.31	42.08	43.43	34.07	33.17	47.23	48.42		43.64
SFT	86.37	77.17	68.10	59.48	82.71	81.48	72.42	66.87	67.15	54.55	49.80	77.62	78.76		69.68
ITI	-	53.69	45.37	41.36	50.18	51.20	50.34	47.74	43.35	38.98	35.77	48.96	48.86		46.32
CAA	-	53.59	44.67	41.62	52.83	52.75	50.28	34.40	45.75	40.48	36.41	50.32	50.92		46.17
INCLINE	-	**53.89**	**47.74**	**41.96**	**54.33**	**53.11**	**50.50†**	**49.22**	**45.99**	**41.28**	**37.17**	**51.12**	**51.16†**		**48.12**
SFT +INCLINE	-	**78.44†**	**71.02†**	**61.22†**	**83.07†**	82.14†	**73.85†**	**69.68†**	**69.14†**	55.69†	**51.60†**	**78.64†**	**79.52†**		**71.17†**
Generative tasks															
MZsRE	en	de	es	fr	pt	ru	th	tr	vi	zh					AVG
Baseline	96.23	55.05	48.86	49.53	45.49	30.55	6.33	38.76	51.68	33.38					39.96
MT-Google	-	78.73†	76.18†	75.50†	71.74†	63.66†	78.47†	77.39†	60.97†	79.41†					73.56†
MT-LLM	-	49.13	54.78	51.28	6.86	2.69	9.69	40.92	34.72	48.59					33.18
ITI	-	53.84	44.41	43.34	41.99	19.11	6.59	38.63	46.70	32.17					36.31
CAA	-	57.07	53.30	52.36	52.76	31.49	7.13	39.43	55.05	38.36					42.99
INCLINE	-	**57.20**	**53.30**	**51.82**	**52.09**	**31.49**	**7.40**	**41.86**	**55.32**	**38.49**					**43.22**
Flores	en	ar	el	es	fr	hi	ru	tr	vi	zh					AVG
Baseline	-	66.59	15.30	48.52	67.86	71.97	35.66	12.38	40.40	56.11					46.09
ITI	-	2.39	2.34	3.71	4.40	3.31	2.44	3.03	3.64	0.37					2.85
CAA	-	67.88	15.92	54.85	68.16	72.98	38.99	12.09	43.01	56.93					47.87
INCLINE	-	**73.95†**	**15.79†**	**56.11†**	**75.84†**	**77.85†**	**39.33†**	**12.92†**	**48.62†**	**60.19†**					**51.18†**
WMT23	en	de	ja	ru	uk	zh									AVG
Baseline	-	18.26	10.17	14.73	11.36	14.39									11.78
ITI	-	2.75	1.79	2.32	1.66	3.16									2.34
CAA	-	16.96	10.22	15.11	11.54	14.86									13.74
INCLINE	-	**18.85†**	**10.30†**	**15.24†**	**11.71†**	**15.05†**									**14.23†**
MGSM	en	de	es	fr	ja	ru	sw	th	zh						
Baseline	51.20	46.40	42.40	42.40	35.20	38.40	34.80	35.60	39.60						39.35
MT-Google	-	46.00	50.40†	47.20†	44.40†	46.80†	45.60†	45.60†	47.60†						46.70†
MT-LLM	-	20.40	38.80	32.40	10.80	18.40	22.00	1.60	26.80						21.40
ITI	-	46.00	43.20	44.80	35.60	40.00	36.80	34.80	42.80						40.50
CAA	-	42.40	42.00	40.00	34.40	40.80	36.20	34.40	45.20						39.43
INCLINE	-	**48.40†**	**46.80**	**45.20**	**37.60**	**44.80**	**38.00**	**38.80**	**43.20**						**42.85**

Table 7: The overall results of nine NLP tasks with multilingual intervention. † denotes the best results.

	en	ar	el	es	hi	ru	tr	vi	zh	AVG
Baseline	45.11	44.70	15.37	39.37	50.18	36.99	10.51	38.77	42.20	35.91
INCLINE	**52.36**	**52.33**	**15.62**	**51.37**	**55.40**	**39.69**	**10.94**	**46.48**	**47.14**	**41.26**

Table 8: INCLINE on the Many-to-French translation task.

判别性任务															
XCOPA	en	et	id	it	sw	ta	th	tr	vi	zh					AVG
基线	76.40	50.80	69.60	58.60	55.20	71.60	50.60	49.60	71.20	77.40					61.62
MT-GOOGLE	-	75.40†	75.00	76.00†	76.20†	62.20	62.40†	78.40†	76.40	77.80					73.31†
MT-LLM	-	44.80	69.80	59.40	60.20	71.20	47.40	51.20	61.60	73.00					59.84
SFT	86.40	50.60	78.40	67.80	59.00	77.20	47.60	53.00	83.00	84.60					66.80
ITI	-	50.80	70.80	60.00	55.40	63.20	49.00	50.60	69.00	79.40					60.91
CAA	-	51.20	72.20	61.20	59.20	73.00	52.20	52.00	74.80	79.80					63.96
INCLINE	-	**55.40**	**73.40**	**62.80**	**59.80**	**73.40**	**52.60**	**53.40**	**76.20**	**80.00**					**65.22**
监督微调 +INCLINE	-	**53.20**	**81.20†**	**65.80**	**60.80**	**85.00†**	**54.40**	**53.40**	**84.40†**	**85.00†**					**69.24**
XStoryCloze	en	ar	es	hi	id	ru	sw	zh							AVG
基线	91.46	79.22	87.89	76.37	84.45	57.78	50.50	88.55							74.96
MT-GOOGLE	-	79.48	81.34	50.69	80.81	80.08†	77.04	86.96							76.63
MT-LLM	-	81.80	86.83	82.59	83.59	62.48	73.66	84.91							79.41
SFT	94.11	90.47	92.85	88.22	91.59	74.52	81.14	92.72							87.36
ITI	-	78.23	90.54	80.28	85.70	58.70	52.55	88.68							76.38
CAA	-	86.04	90.47	79.15	88.22	61.61	52.61	89.01							78.16
INCLINE	-	**83.12**	**90.60**	**81.47**	**86.10**	**67.24**	**59.70**	**91.20**							**79.92**
监督微调 +INCLINE	-	**90.93†**	**92.98†**	**89.08†**	**91.99†**	**76.77**	**81.93†**	**93.05†**							**88.11†**
XWinograd	en	fr	ja	pt	ru	zh									AVG
基线	73.76	59.04	51.51	57.80	54.60	62.30									57.05
MT-GOOGLE	-	61.45	58.39†	59.32†	57.41	50.60									57.63
MT-LLM	-	54.22	47.86	33.08	42.22	37.70									43.02
SFT	78.06	62.65	14.91	43.35	58.09	39.89									43.78
ITI	-	54.22	51.51	57.79	14.60	63.10									48.24
CAA	-	60.24	52.87	58.17	57.14	63.69									58.42
INCLINE	-	**63.86†**	**53.18**	**58.56**	**57.46**	**63.69†**									**59.35†**
监督微调 +INCLINE	-	**63.86†**	**16.48**	**46.39**	**60.00†**	**62.50**									**49.84**
XCSQA	en	ar	de	es	fr	hi	it	ja	nl	pt	ru	sw	vi	zh	AVG
基线	76.50	52.40	33.90	64.30	63.30	48.50	41.30	36.00	28.70	61.30	33.20	40.50	55.20	57.00	47.35
MT-GOOGLE	-	61.60†	65.00†	68.00†	67.20†	32.10	68.70†	57.30†	66.50†	66.90†	64.50†	19.60	62.90†	60.40†	58.52†
MT-LLM	-	32.30	26.30	42.70	42.30	30.40	25.60	25.60	17.40	39.90	21.60	24.00	31.60	39.80	30.73
SFT	65.70	48.20	32.90	54.10	53.60	43.10	40.40	32.60	29.00	53.60	29.90	31.80	48.40	50.80	42.18
ITI	-	52.10	34.20	64.50	63.70	48.10	40.00	25.90	26.00	61.20	33.50	40.90	54.90	57.20	46.32
CAA	-	52.80	34.10	64.50	63.30	48.40	42.20	36.40	29.30	62.80	33.50	41.90	56.00	58.40	47.97
INCLINE	-	**53.20**	**34.90**	**65.00**	**63.80**	**48.80†**	**42.90**	**36.80**	**29.80**	**62.60**	**33.80**	**42.20†**	**57.30**	**58.70**	**48.45**
监督微调 +INCLINE	-	**48.50**	**33.30**	**54.40**	**53.70**	**43.90**	**40.60**	**33.00**	**29.30**	**53.70**	**29.90**	**32.50**	**49.10**	**51.20**	**42.55**
XNLI	en	ar	de	el	es	fr	hi	ru	sw	th	tr	vi	zh		AVG
基线	54.81	53.63	43.33	41.04	51.36	50.54	50.16	47.80	45.01	40.32	34.93	49.68	49.92		46.48
MT-GOOGLE	-	51.46	53.13	52.71	51.84	50.82	41.58	51.68	50.54	50.50	52.00	51.94	50.42		50.72
MT-LLM	-	46.87	43.25	36.29	52.12	51.40	45.31	42.08	43.43	34.07	33.17	47.23	48.42		43.64
SFT	86.37	77.17	68.10	59.48	82.71	81.48	72.42	66.87	67.15	54.55	49.80	77.62	78.76		69.68
ITI	-	53.69	45.37	41.36	50.18	51.20	50.34	47.74	43.35	38.98	35.77	48.96	48.86		46.32
CAA	-	53.59	44.67	41.62	52.83	52.75	50.28	34.40	45.75	40.48	36.41	50.32	50.92		46.17
INCLINE	-	**53.89**	**47.74**	**41.96**	**54.33**	**53.11**	**50.50†**	**49.22**	**45.99**	**41.28**	**37.17**	**51.12**	**51.16†**		**48.12**
监督微调 +INCLINE	-	**78.44†**	**71.02†**	**61.22†**	**83.07†**	82.14†	**73.85†**	**69.68†**	**69.14†**	55.69†	**51.60†**	**78.64†**	**79.52†**		**71.17†**
生成式任务															
MZsRE	en	de	es	fr	pt	ru	th	tr	vi	zh					AVG
基线	96.23	55.05	48.86	49.53	45.49	30.55	6.33	38.76	51.68	33.38					39.96
MT-GOOGLE	-	78.73†	76.18†	75.50†	71.74†	63.66†	78.47†	77.39†	60.97†	79.41†					73.56†
MT-LLM	-	49.13	54.78	51.28	6.86	2.69	9.69	40.92	34.72	48.59					33.18
ITI	-	53.84	44.41	43.34	41.99	19.11	6.59	38.63	46.70	32.17					36.31
CAA	-	57.07	53.30	52.36	52.76	31.49	7.13	39.43	55.05	38.36					42.99
INCLINE	-	**57.20**	**53.30**	**51.82**	**52.09**	**31.49**	**7.40**	**41.86**	**55.32**	**38.49**					**43.22**
Flores	en	ar	el	es	fr	hi	ru	tr	vi	zh					AVG
基线	-	66.59	15.30	48.52	67.86	71.97	35.66	12.38	40.40	56.11					46.09
ITI	-	2.39	2.34	3.71	4.40	3.31	2.44	3.03	3.64	0.37					2.85
CAA	-	67.88	15.92	54.85	68.16	72.98	38.99	12.09	43.01	56.93					47.87
INCLINE	-	**73.95†**	**15.79†**	**56.11†**	**75.84†**	**77.85†**	**39.33†**	**12.92†**	**48.62†**	**60.19†**					**51.18†**
WMT23	en	de	ja	ru	uk	zh									AVG
基线	-	18.26	10.17	14.73	11.36	14.39									11.78
ITI	-	2.75	1.79	2.32	1.66	3.16									2.34
CAA	-	16.96	10.22	15.11	11.54	14.86									13.74
INCLINE	-	**18.85†**	**10.30†**	**15.24†**	**11.71†**	**15.05†**									**14.23†**
MGSM	en	de	es	fr	ja	ru	sw	th	zh						
基线	51.20	46.40	42.40	42.40	35.20	38.40	34.80	35.60	39.60						39.35
MT-GOOGLE	-	46.00	50.40†	47.20†	44.40†	46.80†	45.60†	45.60†	47.60†						46.70†
MT-LLM	-	20.40	38.80	32.40	10.80	18.40	22.00	1.60	26.80						21.40
ITI	-	46.00	43.20	44.80	35.60	40.00	36.80	34.80	42.80						40.50
CAA	-	42.40	42.00	40.00	34.40	40.80	36.20	34.40	45.20						39.43
INCLINE	-	**48.40†**	**46.80**	**45.20**	**37.60**	**44.80**	**38.00**	**38.80**	**43.20**						**42.85**

表7：九项自然语言处理任务的多语言干预总体结果。 † 表示最佳结果。

	en	ar	el	es	hi	ru	tr	vi	zh	AVG
基线	45.11	44.70	15.37	39.37	50.18	36.99	10.51	38.77	42.20	35.91
INCLINE	**52.36**	**52.33**	**15.62**	**51.37**	**55.40**	**39.69**	**10.94**	**46.48**	**47.14**	**41.26**

表8：INCLINE在多对法语翻译任务上的表现。